

Address wording issues related to ISO 10646

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Motivation

Review of some editorial fixes following the recent update of the normative reference to ISO 10646 has unearthed a series of wording issues around the subject. This paper intends to fix those issues by rewording relevant paragraphs.

Proposal

This paper addresses all of the following issues:

1. The current wording in [lex.charset] does not specify what the behaviour is for a universal-character-name without a corresponding short identifier in ISO 10646.

For example, `\U99004141` and `\U00110000`. Neither of these designates a code point in ISO 10646, but the standard is silent about this, which makes the behaviour undefined by omission.

This paper addresses this by making such uses ill-formed, maintaining consistency with the current treatment of surrogate values (`\U0000D800` is already ill-formed).

2. The current wording in [lex.charset] uses “hexadecimal value”, which is confusing because a value is just a number, and hexadecimal is just a way to represent numbers; “value” alone should suffice.

This paper addresses this by removing the need for this term.

3. There is some interest in using the U+ notation (as in U+0041 or U+1F34A) to refer to Unicode code points across the entire standard.

This paper changes all the relevant wording to use U+ notation.

4. The current text includes explanations of terms from ISO 10646 (like “surrogate code point” or “control character”) in normative text, which is undesirable.

This paper moves such explanations to non-normative text, and clarifies some existing explanations.

Technical Specifications

In this description, text that should be deleted is marked red and striked out; text that should be added is marked green and underlined. Apply these changes on top of the editorial fix provided in [PR #2201](https://github.com/cplusplus/draft/pull/2201) (<https://github.com/cplusplus/draft/pull/2201>).

Edit 5.3 [lex.charset], paragraph 2 as follows.

² The *universal-character-name* construct provides a way to name other characters.

hex-quad: *hexadecimal-digit hexadecimal-digit hexadecimal-digit hexadecimal-digit*

universal-character-name: *

Edit 5.13.3 [lex.ccon], paragraph 3 as follows.

³ A character literal that begins with `u8`, such as `u8 'w'`, is a character literal of type `char`, known as a *UTF-8 character literal*. The value of a UTF-8 character literal is equal to its ISO 10646 code point value, provided that the code point value is representable with a single UTF-8 code unit (~~that is, provided it is in the C0 Controls and Basic Latin Unicode block~~)[~~Note—~~that is, provided it is in the range 0x0-0x7F, inclusive—end note]. If the value is not representable with a single UTF-8 code unit, the program is ill-formed. A UTF-8 character literal containing multiple *c-chars* is ill-formed.

Edit 5.13.3 [lex.ccon], paragraph 4 as follows.

⁴ A character literal that begins with the letter `u`, such as `u 'x'`, is a character literal of type `char16_t`. The value of a `char16_t` character literal containing a single *c-char* is equal to its ISO 10646 code point value, provided that the code point value is representable with a single 16-bit code unit (~~[Note—~~that is, provided it is in the basic multi-lingual plane the range 0x0-0xFFFF, inclusive)—end note]. If the value is not representable with a single 16-bit code unit, the program is ill-formed. A `char16_t` character literal containing multiple *c-chars* is ill-formed.

Edit 5.13.3 [lex.string], paragraph 10 as follows.

¹⁰ A *string-literal* that begins with `u`, such as `u"asdf"`, is a `char16_t` string literal. A `char16_t` string literal has type “array of *n* `const char16_t`”, where *n* is the size of the string as defined below; it is initialized with the given characters. A single *c-char* may produce more than one `char16_t` character in the form of surrogate pairs [~~Note—~~a surrogate pair is a representation for a single character as a sequence of two 16-bit code units—end note].

Edit 19.8 [cpp.predefined], item (2.4) as follows.

(2.4) `__STDC_ISO_10646__`

An integer literal of the form `yyyymmL` (for example, `199712L`). If this symbol is defined, then every character in the Unicode required set, when stored in an object of type `wchar_t`, has the same value as the ~~short identifier~~code point of that character. The Unicode required set consists of all the characters that are defined by ISO/IEC 10646, along with all amendments and technical corrigenda as of the specified year and month.